

http

No.	Time	Source	Destination	Protocol	Length	Info
199	5.297341	192.168.1.102	128.119.245.12	HTTP	104	POST /ethereal-labs/lab3-1-reply.htm HTTP/1.1 (text/plain)
203	5.461175	128.119.245.12	192.168.1.102	HTTP	784	HTTP/1.1 200 OK (text/html)

Wireshark · Packet 199 · tcp-ethereal-trace-1

Transmission Control Protocol, Src Port: 1161, Dst Port: 80, Seq: 164041, Ack: 1, Len: 50

Source Port: 1161

Destination Port: 80

[Stream index: 0]

[Stream Packet Number: 196]

[Conversation completeness: Incomplete, DATA (15)]

[TCP Segment Len: 50]

Sequence Number: 164041 (relative sequence number)

Sequence Number (raw): 232293053

[Next Sequence Number: 164091 (relative sequence number)]

Acknowledgment Number: 1 (relative ack number)

Acknowledgment number (raw): 883061786

0101 = Header Length: 20 bytes (5)

Flags: 0x018 (PSH, ACK)

Window: 17520

[Calculated window size: 17520]

[Window size scaling factor: -2 (no window scaling used)]

Checksum: 0x9f0f [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

[Timestamps]

[SEQ/ACK analysis]

TCP payload (50 bytes)

TCP segment data (50 bytes)

[122 Reassembled TCP Segments (164090 bytes): #4(565), #5(1460), #7(1460), #8(1460), #10(1460), #11(1460), #13(1147), #18(1460), #19(1460), #20(1460), #21(1460), #

0000 00 06 25 da af 73 00 20 e0 8a 70 1a 08 00 45 00 ..%..s...p...E

0010 00 5a 1e 9a 40 00 80 06 a4 71 c0 a8 01 66 80 77 .Z...@...q...f.w

0020 f5 0c 04 89 00 50 0d d8 82 bd 34 a2 74 1a 50 18P...4.t.P

0030 44 70 9f 0f 00 00 0d 0a 2d 2d 2d 2d 2d 2d 2d 2d Dp.....

0040 2d 2d 2d 2d 2d 2d 2d 2d 2d 2d 2d 2d 2d 2d 2d 2d

Frame (104 bytes) Reassembled TCP (164090 bytes)

No: 199 · Time: 5.297341 · Source: 192.168.1.102 · Destination: 128.119.245.12 · Protocol: HTTP · Length: 104 · Info: POST /ethereal-labs/lab3-1-reply.htm HTTP/1.1 (text/plain)

Show packet bytes

Layout: Vertical (Stacked)

從 HTTP 包做判斷: client 會先跟 server 通信並執行

1. Client IP address: 192.168.1.102

TCP port number: 1161 #

2. Server IP address: 128.119.245.12

TCP port number: 80 預設 #

3. 由於使用的為別人執行的記錄檔

並非使用自己 computer.

IP: TCP 與 Q1 相同.

IP address is 192.168.1.102. port number is 1161 #

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.102	128.119.245.12	TCP	62	1161 → 80 [SYN] Seq=0 Win=16384 Len=0 MSS=1460 SACK_PERM
2	0.023172	128.119.245.12	192.168.1.102	TCP	62	80 → 1161 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460 SACK_PERM

Frame 1: 62 bytes on wire (496 bits), 62 bytes captured (496 bits)
 Ethernet II, Src: ActiontecEle_8a:70:1a (00:20:e0:8a:70:1a), Dst: LinksysGroup_da:af:73 (00:06:25:da:af:73)
 Internet Protocol Version 4, Src: 192.168.1.102, Dst: 128.119.245.12
 Transmission Control Protocol, Src Port: 1161, Dst Port: 80, Seq: 0, Len: 0

Source Port: 1161
 Destination Port: 80
 [Stream index: 0]
 [Stream Packet Number: 1]
 [Conversation completeness: Incomplete, DATA (15)]
 [TCP Segment Len: 0]
 Sequence Number: 0 (relative sequence number)
 Sequence Number (raw): 232129012
 [Next Sequence Number: 1 (relative sequence number)]
 Acknowledgment Number: 0
 Acknowledgment number (raw): 0
 0111 = Header Length: 28 bytes (7)
 Flags: 0x002 (SYN)

000. = Reserved: Not set
 ...0 = Accurate ECN: Not set
 0... = Congestion Window Reduced: Not set
 0... = ECN-Echo: Not set
0. = Urgent: Not set
0 = Acknowledgment: Not set
0 = Push: Not set
0 = Reset: Not set
1 = Syn: Set
0 = Fin: Not set
 [TCP Flags:S-]
 Window: 16384
 [Calculated window size: 16384]
 Checksum: 0xf6e9 [unverified]
 [Checksum Status: Unverified]
 Urgent Pointer: 0
 Options: (8 bytes), Maximum segment size, No-Operation (NOP), No-Operation (NOP), SACK permitted
 [Timestamps]

pkt 1 relative sequence number is 0
 raw sequence number is 232129012
 4. 同於 client 與 server 連結
 的 TCP SYN segment is pkt 1
 1. 為 client 與 server 端 connect
 常 SYN flag set 1
 可以判斷其為 SYN segment

This shows the raw value of the sequence number (tcp.seq_raw), 4 byte(s)

Wireshark · Packet 2 · tcp-ethereal-trace-1

Transmission Control Protocol, Src Port: 80, Dst Port: 1161, Seq: 0, Ack: 1, Len: 0

Source Port: 80
 Destination Port: 1161
 [Stream index: 0]
 [Stream Packet Number: 2]
 [Conversation completeness: Incomplete, DATA (15)]
 [TCP Segment Len: 0]
 Sequence Number: 0 (relative sequence number)
 Sequence Number (raw): 883061785
 [Next Sequence Number: 1 (relative sequence number)]
 Acknowledgment Number: 1 (relative ack number)
 Acknowledgment number (raw): 232129013
 0111 = Header Length: 28 bytes (7)
 Flags: 0x012 (SYN, ACK)

000. = Reserved: Not set
 ...0 = Accurate ECN: Not set
 0... = Congestion Window Reduced: Not set
0. = ECN-Echo: Not set
0. = Urgent: Not set
1 = Acknowledgment: Set
0 = Push: Not set
0 = Reset: Not set
1 = Syn: Set
0 = Fin: Not set
 [TCP Flags:A..S-]
 Window: 5840

pkt 2 relative sequence number is 0
 raw sequence number is 883061785
 5. pkt 2 為 SYNACK segment
 在 server 端發送回 client 的
 接收到的建立上連接
 ACK SYN
 value of Ack field in SYNACK segment
 is 1 (relative)
 raw ack number is 232129013
 server 端由 client 的 sequence number + 1
 決定 ACK number in SYNACK segment
 常 SYN flag 與 ACK flag set 1
 可以得知 packet 為 SYNACK segment

No.: 2 · Time: 0.023172 · Source: 128.119.245.12 · Destination: 192.168.1.102 · Protocol: TCP · Length: 62 · Info: 80 → 1161 [SYN, ACK] Seq=0 Ack=1 Win=5840
 Show packet bytes Layout: Vertical (Stacked)

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.102	128.119.245.12	TCP	62	1161 → 80 [SYN] Seq=0 Win=16384 Len=0 MSS=1460 SACK_PERM
2	0.023172	128.119.245.12	192.168.1.102	TCP	62	80 → 1161 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460 SACK_PERM
3	0.023265	192.168.1.102	128.119.245.12	TCP	54	1161 → 80 [ACK] Seq=1 Ack=1 Win=17520 Len=0
4	0.026477	192.168.1.102	128.119.245.12	TCP	619	1161 → 80 [PSH, ACK] Seq=1 Ack=1 Win=17520 Len=565 [TCP PDU reassembled in 199]

Source Port: 1161
Destination Port: 80
[Stream index: 0]
[Stream Packet Number: 4]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 565]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 232129013
[Next Sequence Number: 566 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 883061786
0101 = Header Length: 20 bytes (5)
Flags: 0x018 (PSH, ACK)
000. = Reserved: Not set
...0 = Accurate ECN: Not set
.... 0... = Congestion Window Reduced: Not set
.... 0... = ECN-Echo: Not set
.... ..0. = Urgent: Not set
.... ...1 = Acknowledgment: Set
.... 1... = Push: Set
....0.. = Reset: Not set
....0. = Syn: Not set
....0 = Fin: Not set
[TCP Flags:AP...]
Window: 17520
[Calculated window size: 17520]
[Window size scaling factor: -2 (no window scaling used)]
Checksum: 0x1fbd [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[SEQ/ACK analysis]
TCP payload (565 bytes)
[Reassembled PDU in frame: 199]
TCP segment data (565 bytes)

b. pkt 4 containing the HTTP Post command
the value of sequence number is 1 (relative)
raw sequence number is 232129013

232129012 + 1

#

0010	02 5d 1e 21 40 00 80 06	a2 e7 c0 a8 01 66 80 77	..]!@... ..f.w
0020	f5 0c 04 89 00 50 0d d6	01 f5 34 a2 74 1a 50 18	P...4..t.P.
0030	44 70 1f bd 00 00 50 4f	53 54 20 2f 65 74 68 65	Dp...PO ST /the
0040	72 65 61 6c 2d 6c 61 62	73 2f 6c 61 62 33 2d 31	real-lab-3/lab3-1
0050	2d 72 65 70 6c 79 2e 68	74 6d 20 48 54 54 50 2f	-reply.htm HTTP/
0060	31 2e 31 0d 0a 48 6f 73	74 3a 20 67 61 69 61 2e	1.1..Hos t: gaia.
0070	63 73 2e 75 6d 61 73 73	2e 65 64 75 0d 0a 55 73	cs.umass .edu..Us
0080	65 72 2d 41 67 65 6e 74	3a 20 4d 6f 7a 69 6c 6c	er-Agent : Mozill
0090	61 2f 35 2e 30 20 28 57	69 6e 64 6f 77 73 3b 20	a/5.0 (W indows;
00a0	55 3b 20 57 69 6e 64 6f	77 73 20 4e 54 20 35 2e	U; Windo ws NT 5.
00b0	31 3b 20 65 6e 2d 55 53	3b 20 72 76 3a 31 2e 30	1; en-US ; rv:1.0
00c0	2e 32 29 20 47 65 63 6b	6f 2f 32 30 30 33 30 32	.2) Geck o/200302
00d0	30 38 20 4e 65 74 73 63	61 70 65 2f 37 2e 30 32	08 Netsc ape/7.02
00e0	0d 0a 41 63 63 65 70 74	3a 20 74 65 78 74 2f 78	..Accept : text/x
00f0	6d 6c 2c 61 70 70 6c 69	63 61 74 69 6f 6e 2f 78	ml,appli cation/x
0100	6d 6c 2c 61 70 70 6c 69	63 61 74 69 6f 6e 2f 78	ml,appli cation/x
0110	68 74 6d 6c 2b 78 6d 6c	2c 74 65 78 74 2f 68 74	html+xml ,text/ht
0120	6d 6c 3b 71 3d 30 2e 39	2c 74 65 78 74 2f 70 6c	ml;q=0.9 ,text/pl
0130	61 69 6e 3b 71 3d 30 2e	38 2c 76 69 64 65 6f 2f	ain;q=0. 8,video/
0140	78 2d 6d 6e 67 2c 69 6d	61 67 65 2f 70 6e 67 2c	x-mng,im age/png,
0150	69 6d 61 67 65 2f 6a 70	65 67 2c 69 6d 61 67 65	image/jp eg,image
0160	2f 67 69 66 3b 71 3d 30	2e 32 2c 74 65 78 74 2f	/gif;q=0. 2,text/
0170	63 73 73 2c 2a 2f 2a 3b	71 3d 30 2e 31 0d 0a 41	css,*/*; q=0.1..A
0180	63 63 65 70 74 2d 4c 61	6e 67 75 61 67 65 3a 20	ccept-La nguage:
0190	65 6e 2d 75 73 2c 20 65	6e 3b 71 3d 30 2e 35 30	en-us, e n;q=0.50
01a0	0d 0a 41 63 63 65 70 74	2d 45 6e 63 6f 64 69 6e	..Accept -Encodin
01b0	67 3a 20 67 7a 69 70 2c	20 64 65 66 6c 61 74 65	g: gzip, deflate
01c0	2c 20 63 6f 6d 70 72 65	73 73 3b 71 3d 30 2e 39	, compre ss;q=0.9
01d0	0d 0a 41 63 63 65 70 74	2d 43 68 61 72 73 65 74	..Accept -Charset
01e0	3a 20 49 53 4f 2d 38 38	35 39 2d 31 2c 20 75 74	: ISO-88 59-1, ut
01f0	66 2d 38 3b 71 3d 30 2e	36 36 2c 20 2a 3b 71 3d	f-8;q=0. 66, *;q=
0200	30 2e 36 36 0d 0a 4b 65	65 70 2d 41 6c 69 76 65	0.66..Ke ep-Alive
0210	3a 20 33 30 30 0d 0a 43	6f 6e 6e 65 63 74 69 6f	: 300..C onnectio
0220	6e 3a 20 6b 65 65 70 2d	61 6c 69 76 65 0d 0a 52	n: keep- alive..R
0230	65 6e 65 72 65 72 3a 20	68 74 74 70 3a 2f 2f 67	eferer: http://g
0240	61 69 61 2e 63 73 2e 75	6d 61 73 73 2e 65 64 75	aia.cs.u mass.edu
0250	2f 65 74 68 65 72 65 61	6c 2d 6c 61 62 73 2f 6c	/etherea l-labs/l
0260	61 62 33 2d 31 2e 68 74	6d 0d 0a	ab3-1.ht m..

7.

Q8

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.102	128.119.245.12	TCP	62	1161 → 80 [SYN] Seq=0 Win=16384 Len=0 MSS=1460 SACK_PERM
2	0.023172	128.119.245.12	192.168.1.102	TCP	62	80 → 1161 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460 SACK_PERM
3	0.023265	192.168.1.102	128.119.245.12	TCP	54	1161 → 80 [ACK] Seq=1 Ack=1 Win=17520 Len=0
4	0.026477	192.168.1.102	128.119.245.12	TCP	619	1161 → 80 [PSH, ACK] Seq=1 Ack=1 Win=17520 Len=565 [TCP PDU reassembled in 199]
5	0.041737	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [PSH, ACK] Seq=566 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
6	0.053937	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=566 Win=6780 Len=0 ACK = Seq + Len = 1 + 565
7	0.054026	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=2026 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
8	0.054690	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=3486 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
9	0.077294	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=2026 Win=8760 Len=0
10	0.077405	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=4946 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
11	0.078157	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=6406 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
12	0.124085	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=3486 Win=11680 Len=0
13	0.124185	192.168.1.102	128.119.245.12	TCP	1201	1161 → 80 [PSH, ACK] Seq=7866 Ack=1 Win=17520 Len=1147 [TCP PDU reassembled in 199]
14	0.169118	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=4946 Win=14600 Len=0
15	0.217299	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=6406 Win=17520 Len=0
16	0.267802	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=7866 Win=20440 Len=0
17	0.304807	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=9013 Win=23360 Len=0

8. 除了 pkt 4, length 是 565 bytes 外
其他 5 個 pkt (5, 7 及 10, 11) 皆是 1460 bytes

consider the TCP segment containing HTTP Post as the first segment

the first six segments in the TCP connection is.	pkt	4	5	7	8	10	11
	sequence number	1	566	2026	3486	4946	6406
ACK segment of segment 1~6	pkt	6	9	12	14	15	16

	sent time (sec)	ACK received time (sec)	RTT (sec) = receive - sent time
segment 1	0.026477	0.053937	0.02746
2	0.041737	0.077294	0.035557
3	0.054026	0.124085	0.070059
4	0.054690	0.169118	0.11443
5	0.077405	0.217299	0.13989
6	0.078157	0.267802	0.18964

從 ACK 可以看到更精確 RTT!

$$\text{Estimated RTT} = \frac{0.875 \times \text{estimated RTT} + 0.125 \times \text{sample RTT}}{(1-\alpha)} \quad (\alpha = 0.125)$$

Estimated RTT after receipt ACK of segment 1 = 0.02746 (sec).

$$2. \quad 0.875 \times 0.02746 + 0.125 \times 0.035557 = 0.0285 \text{ (sec).}$$

$$3. \quad 0.875 \times 0.0285 + 0.125 \times 0.070059 = 0.0337$$

4.	$0.875 \times 0.0337 + 0.125 \times 0.11443 \doteq 0.0438$
5.	$0.875 \times 0.0438 + 0.125 \times 0.13989 \doteq 0.0558$
6.	$0.875 \times 0.0558 + 0.125 \times 0.18964 \doteq 0.0725$



Wireshark - Packet 2 - tcp-ethereal-trace-1

可能 server 回傳的第 1 個 ACK 查找

Transmission Control Protocol, Src Port: 80, Dst Port: 1161, Seq: 0, Ack: 1

Source Port: 80
Destination Port: 1161
[Stream index: 0]
[Stream Packet Number: 2]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 883061785
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 232129013
0111 = Header Length: 28 bytes (7)
Flags: 0x012 (SYN, ACK) → minimum amount of available
Window: 5840 → buffer space advertised at the
[Calculated window size: 5840] received for the entire trace is 5840 bytes
Checksum: 0x774d [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (8 bytes), Maximum segment size, No-Operation (NOP), No-Operation
[Timestamps]
[SEQ/ACK analysis]
[This is an ACK to the segment in frame: 1]
[The RTT to ACK the segment was: 0.023172000 seconds]
[iRTT: 0.023265000 seconds]

接收端在 server

0000 00 20 e0 8a 70 1a 00 06 25 da af 73 08 00 45 00 - - - p - - - % - s - - E -
0010 00 30 00 00 40 00 37 06 0c 36 80 77 f5 0c c0 a8 - 0 - @ - 7 - - 6 - w - - -
0020 01 66 00 50 04 89 34 a2 74 19 0d d6 01 f5 70 12 - f - P - - 4 - t - - - - p -
0030 16 d0 77 4d 00 00 02 04 05 b4 01 01 04 02 - - w M - - - - - - - -

No.: 2 - Time: 0.023172 - Source: 128.119.245.12 - Destination: 192.168.1.102 - Protocol: TCP - Length: 62 - Info: 80 → 1161

✓ Show packet bytes Layout: Vertical (Stacked)

[iRTT: 0.023265000 seconds]

tcp-ethereal-trace-1

Wireshark - Packet 202 - tcp-ethereal-trace-1

從最後的 ACK 去判斷

Transmission Control Protocol, Src Port: 80, Dst Port: 1161, Seq: 1, Ack: 164091, Len: 0

Source Port: 80
Destination Port: 1161
[Stream index: 0]
[Stream Packet Number: 199]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 883061786
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 164091 (relative ack number)
Acknowledgment number (raw): 232293103
0101 = Header Length: 20 bytes (5)
Flags: 0x010 (ACK) → max receiver buffer size can grow to 62780 bytes.
Window: 62780 → sender 不會因為 receiver 缺乏 buffer space
[Calculated window size: 62780] 而被限制
[Window size scaling factor: -2 (no window scaling used)]
Checksum: 0x44a8 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[SEQ/ACK analysis]
[This is an ACK to the segment in frame: 199] #
[The RTT to ACK the segment was: 0.158489000 seconds]
[iRTT: 0.023265000 seconds]

0000 00 20 e0 8a 70 1a 00 06 25 da af 73 08 00 45 00 - - - p - - - % - s - - E -
0010 00 28 58 bb 40 00 37 06 b3 82 80 77 f5 0c c0 a8 - (X - @ - 7 - - - w - - -
0020 01 66 00 50 04 89 34 a2 74 1a 0d d8 82 ef 50 10 - f - P - - 4 - t - - - - P -
0030 f5 3c 44 a8 00 00 e5 e7 00 00 07 fb - < D - - - - - - - -

No.: 202 - Time: 5.455830 - Source: 128.119.245.12 - Destination: 192.168.1.102 - Protocol: TCP - Length: 60 - Info: 80 → 1161 [ACK] Seq=1 Ack=164091 Win=62780 Len=0

✓ Show packet bytes Layout: Vertical (Stacked)

Packets: 213 Profile: Default

10. trace file 中沒有 retransmitted segment
(重傳).

如果發生 retransmitt, 則 ACK number 應該會出現重複
的狀態, 表示有前方有 pkt 未接收到



11.

1	0.000000	192.168.1.102	128.119.245.12	TCP	62 1161 → 80 [SYN] Seq=0 Win=16384 Len=0 MSS=1460 SACK_PERM
2	0.023172	128.119.245.12	192.168.1.102	TCP	62 80 → 1161 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460 SACK_PERM
3	0.023265	192.168.1.102	128.119.245.12	TCP	54 1161 → 80 [ACK] Seq=1 Ack=1 Win=17520 Len=0
4	0.026477	192.168.1.102	128.119.245.12	TCP	619 1161 → 80 [PSH, ACK] Seq=1 Ack=1 Win=17520 Len=565 [TCP PDU reassembled in 199]
5	0.041737	192.168.1.102	128.119.245.12	TCP	1514 1161 → 80 [PSH, ACK] Seq=566 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
6	0.053937	128.119.245.12	192.168.1.102	TCP	60 80 → 1161 [ACK] Seq=1 Ack=566 Win=6780 Len=0
7	0.054026	192.168.1.102	128.119.245.12	TCP	1514 1161 → 80 [ACK] Seq=2026 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
8	0.054690	192.168.1.102	128.119.245.12	TCP	1514 1161 → 80 [ACK] Seq=3486 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
9	0.077294	128.119.245.12	192.168.1.102	TCP	60 80 → 1161 [ACK] Seq=1 Ack=2026 Win=8760 Len=0
10	0.077405	192.168.1.102	128.119.245.12	TCP	1514 1161 → 80 [ACK] Seq=4946 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
11	0.078157	192.168.1.102	128.119.245.12	TCP	1514 1161 → 80 [ACK] Seq=6406 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
12	0.124085	128.119.245.12	192.168.1.102	TCP	60 80 → 1161 [ACK] Seq=1 Ack=3486 Win=11680 Len=0
13	0.124185	192.168.1.102	128.119.245.12	TCP	1201 1161 → 80 [PSH, ACK] Seq=7866 Ack=1 Win=17520 Len=1147 [TCP PDU reassembled in 199]
14	0.169118	128.119.245.12	192.168.1.102	TCP	60 80 → 1161 [ACK] Seq=1 Ack=4946 Win=14600 Len=0
15	0.217299	128.119.245.12	192.168.1.102	TCP	60 80 → 1161 [ACK] Seq=1 Ack=6406 Win=17520 Len=0
16	0.267802	128.119.245.12	192.168.1.102	TCP	60 80 → 1161 [ACK] Seq=1 Ack=7866 Win=20440 Len=0
17	0.304807	128.119.245.12	192.168.1.102	TCP	60 80 → 1161 [ACK] Seq=1 Ack=9013 Win=23360 Len=0
18	0.305040	192.168.1.102	128.119.245.12	TCP	1514 1161 → 80 [ACK] Seq=9013 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
19	0.305813	192.168.1.102	128.119.245.12	TCP	1514 1161 → 80 [ACK] Seq=10473 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
20	0.306692	192.168.1.102	128.119.245.12	TCP	1514 1161 → 80 [ACK] Seq=11933 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
21	0.307571	192.168.1.102	128.119.245.12	TCP	1514 1161 → 80 [ACK] Seq=13393 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
22	0.308699	192.168.1.102	128.119.245.12	TCP	1514 1161 → 80 [ACK] Seq=14853 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]

一般來說 reciever 在 ACK 中 確認 1460 bytes. (偶爾會有不同 566, 892, 1147 bytes 等等)

No.	Time	Source	Destination	Protocol	Length	Info
68	1.394202	192.168.1.102	128.119.245.12	TCP	946	1161 → 80 [PSH, ACK] Seq=49081 Ack=1 Win=17520 Len=892 [TCP PDU reassembled in 199]
69	1.488313	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=44701 Win=62780 Len=0
70	1.584980	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=47621 Win=62780 Len=0
71	1.661513	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=49973 Win=62780 Len=0
72	1.661734	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=49973 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
73	1.662474	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=51433 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
74	1.663315	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=52893 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
75	1.664198	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=54353 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
76	1.665254	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=55813 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
77	1.666151	192.168.1.102	128.119.245.12	TCP	946	1161 → 80 [PSH, ACK] Seq=57273 Ack=1 Win=17520 Len=892 [TCP PDU reassembled in 199]
78	1.758227	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=52893 Win=62780 Len=0
79	1.860063	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=55813 Win=62780 Len=0
80	1.930880	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=58165 Win=62780 Len=0
81	1.931099	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=58165 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
82	1.931879	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=59625 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
83	1.932757	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=61085 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
84	1.933636	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=62545 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
85	1.934770	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=64005 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
86	1.935586	192.168.1.102	128.119.245.12	TCP	946	1161 → 80 [PSH, ACK] Seq=65465 Ack=1 Win=17520 Len=892 [TCP PDU reassembled in 199]
87	2.029069	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=61085 Win=62780 Len=0
88	2.126682	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=64005 Win=62780 Len=0
89	2.203195	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=66357 Win=62780 Len=0

There are cases where reciever is ACKing every other segment.

沒收到前面的 ACK, 但有收到後面的 ACK, 表示 pkt 都有傳到.

會發生此情況可能是因為

(計時過早結束 or ACK 丟失等等)

如 pkt 79~80 中 就可以發現 ACK 加了 2352, 為 1460 + 892.

又如 pkt 87~88 中 ACK 加了 2920, 為 1460 × 2.



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.102	128.119.245.12	TCP	62	1161 → 80 [SYN] Seq=0 Win=16384 Len=0 MSS=1460 SACK_PERM
2	0.023172	128.119.245.12	192.168.1.102	TCP	62	80 → 1161 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460 SACK_PERM
3	0.023265	192.168.1.102	128.119.245.12	TCP	54	1161 → 80 [ACK] Seq=1 Ack=1 Win=17520 Len=0
4	0.026477	192.168.1.102	128.119.245.12	TCP	619	1161 → 80 [PSH, ACK] Seq=1 Ack=1 Win=17520 Len=565 [TCP PDU reassembled in 199]

Source Port: 1161	0010 02 5d 1e 21 40 00 80 06 a2 e7 c0 a8 01 66 80 77	] l0 f.w
Destination Port: 80	0020 f5 0c 04 89 00 50 0d d6 01 f5 34 a2 74 1a 50 18	... P... .. 4 t P
[Stream index: 0]	0030 44 70 1f bd 00 00 50 4f 53 54 20 2f 65 74 68 69	Dp... P0 ST /ethe
[Stream Packet Number: 4]	0040 72 65 61 6c 2d 6c 61 62 73 2f 6c 61 62 33 2d 31	real-lab s/lab3-1
[Conversation completeness: Incomplete, DATA (15)]	0050 2d 72 65 70 6c 79 2e 68 74 6d 20 48 54 54 50 2f	-reply.h tm HTTP/
[TCP Segment Len: 565]	0060 31 2e 31 0d 0a 48 6f 73 74 3a 20 67 61 69 61 2e	1.1-Hos t: gaia.
Sequence Number: 1 (relative sequence number)	0070 63 73 2e 75 6d 61 73 73 2e 65 64 75 0d 0a 55 73	cs.umass .edu-Us
Sequence Number (raw): 232129013	0080 65 72 2d 41 67 65 6e 74 3a 20 4d 6f 74 69 6c 6c	er-Agent : Mozill
[Next Sequence Number: 1 (relative sequence number)]	0090 61 2f 35 2e 30 20 28 57 69 6e 64 6f 77 73 3b 20	a/5.0 (W ndows;
Acknowledgment Number: 1 (relative ack number)	00a0 55 3b 20 57 69 6e 64 6f 77 73 20 4e 54 20 35 2e	U; Windo ws nt 5.
Acknowledgment number (raw): 883061786	00b0 31 3b 20 65 6e 2d 55 53 3b 20 72 76 3a 31 2e 30	1; en-US ; rv:1.0
0101 = Header Length: 20 bytes (5)	00c0 2e 32 29 20 47 65 63 6b 6f 2f 32 30 30 33 30 32	.2) Geck o/200302
Flags: 0x018 (PSH, ACK)	00d0 30 38 20 4e 65 74 73 63 61 70 65 2f 37 2e 30 32	08 Netsc ape/7.02
000. = Reserved: Not set	00e0 0d 0a 41 63 63 65 70 74 3a 20 74 65 78 74 2f 78	..Accept : text/x
...0. = Accurate ECN: Not set	00f0 6d 6c 2c 61 70 70 6c 69 63 61 74 69 6f 6e 2f 78	ml,appli cation/x
...0. = Congestion Window Reduced: Not set	0100 6d 6c 2c 61 70 70 6c 69 63 61 74 69 6f 6e 2f 78	ml,appli cation/x
...0. = ECN-Echo: Not set	0110 68 74 6d 6c 2b 78 6d 6c 2c 74 65 78 74 2f 68 74	html+xml;text/ht
...0. = Urgent: Not set	0120 6d 6c 3b 71 3d 30 2e 39 2c 74 65 78 74 2f 70 6c	ml;q=0.9;text/pl
...1. = Acknowledgment: Set	0130 61 69 6e 3b 71 3d 30 2e 38 2c 76 69 64 65 6f 2f	an;q=0.8;video/
...1. = Reset: Set	0140 78 2d 6d 6e 67 2c 69 6d 61 67 65 2f 70 6e 67 2d	x-mng;im age/png;
...0. = Push: Set	0150 69 6d 61 67 65 2f 6a 70 65 67 2c 69 6d 61 67 65	image/jp eg,image
...0. = Sym: Not set	0160 2f 67 69 66 3b 71 3d 30 2e 32 2c 74 65 68 74 2f	/gif;q=0.2;text/
...0. = Fin: Not set	0170 63 73 73 2c 2a 2f 2a 3b 71 3d 30 2e 31 0d 0a 41	css;/*; q=0.1-A
[TCP Flags:AP...]	0180 63 63 65 70 74 2d 4c 61 6e 67 7f 61 67 65 3a 20	ccept-La nguage:
Window: 17520	0190 65 6e 2d 75 73 2c 20 65 6e 3b 71 3d 30 2e 35 30	en-us, e n;q=0.50
[Calculated window size: 17520]	01a0 0d 0a 41 63 63 65 70 74 2d 45 6e 63 6f 64 69 6e	..Accept -encodin
[Window size scaling factor: -2 (no window scaling used)]	01b0 67 3a 20 67 7a 69 70 2c 20 64 65 66 6c 61 74 65	g: gzip, deflate
Checksum: 0xfdb [unverified]	01c0 2c 20 63 6f 6d 70 72 65 73 73 3b 71 3d 30 2e 39	, compr ss;q=0.9
Checksum Status: Unverified	01d0 0d 0a 41 63 63 65 70 74 2d 43 68 61 72 73 65 74	..Accept -Charset
Urgent Pointer: 0	01e0 3a 20 49 53 4f 4d 38 38 35 39 2d 31 2c 20 75 74	: ISO-88 59-1, ut
[Timestamps]	01f0 66 2d 38 3b 71 3d 30 2e 36 3e 2c 20 2a 3b 71 3d	f-8;q=0.66,*;q=
[SEQ/ACK analysis]	0200 30 2e 36 3f 0d 0a 4b 65 65 70 2d 41 6c 69 76 69	0.66-Ke ep-Alive
TCP payload (565 bytes)	0210 3a 20 3a 30 30 0d 0a 43 6f 6e 6e 65 63 74 69 6f	: 300-C onnectio
[Reassembled PDU in frame: 199]	0220 6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65 0d 0a 52	n: keep-alive-R
TCP segment data (565 bytes)	0230 65 6e 65 72 65 72 3a 20 68 74 70 3a 2f 2f 6f	efereen: http://g
	0240 61 69 61 2e 63 73 2e 75 6d 61 73 73 2e 65 64 75	aia.cs.u mass.edu
	0250 2f 65 74 68 65 72 65 61 6c 2d 6c 61 62 73 2f 6e	/etherea l-lab3/l
	0260 61 62 33 2d 31 2e 68 74 6d 0d 0a	ab3-1.ht m...

總傳輸資料量

總傳輸時間

$$= \frac{164091 - 1}{5.45583 - 0.026477} = \frac{30222.754 \sim \text{bytes/sec}}{30.223 \text{ K bytes/sec}}$$

No.	Time	Source	Destination	Protocol	Length	Info
188	5.105060	LinksysGroup da:af:73	Intel 52:2b:23	ARP	42	192.168.1.1 is at 00:06:25:da:af:73
189	5.106121	192.168.1.100	192.168.1.1	SSDP	175	M-SEARCH * HTTP/1.1
190	5.125019	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=154117 Win=62780 Len=0
191	5.197286	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=156469 Win=62780 Len=0
192	5.197508	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=156469 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
193	5.198388	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=157929 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
194	5.199275	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=159389 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
195	5.200252	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=160849 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
196	5.201150	192.168.1.102	128.119.245.12	TCP	1514	1161 → 80 [ACK] Seq=162309 Ack=1 Win=17520 Len=1460 [TCP PDU reassembled in 199]
197	5.202024	192.168.1.102	128.119.245.12	TCP	326	1161 → 80 [PSH, ACK] Seq=163769 Ack=1 Win=17520 Len=272 [TCP PDU reassembled in 199]
198	5.207257	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=159389 Win=62780 Len=0
199	5.297341	192.168.1.102	128.119.245.12	HTTP	144	POST /ethereal-lab3/lab3-1-reply.htm HTTP/1.1 (text/plain)
200	5.389471	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=162309 Win=62780 Len=0
201	5.447887	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=164041 Win=62780 Len=0
202	5.455830	128.119.245.12	192.168.1.102	TCP	60	80 → 1161 [ACK] Seq=1 Ack=164091 Win=62780 Len=0

Ethernet II, Src: LinksysGroup da:af:73 (00:06:25:da:af:73), Dst: AciontecEle 8a:70:1a (00:20:e0:8a:70:1a)	0000 00 20 e0 8a 70 1a 00 06 25 da af 73 08 00 45 00	...p...%..s..E
Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.1.102	0010 00 28 58 bb 40 00 37 06 b3 82 80 77 f5 0c c0 a8	(X @ 7 ...w...
Transmission Control Protocol, Src Port: 80, Dst Port: 1161, Seq: 1, Ack: 164091, Len: 0	0020 01 66 00 50 04 89 34 a2 74 1a 0d d8 82 ef 50 10	f.P..4. t....P
Source Port: 80	0030 f5 3c 4a a8 00 00 e5 e7 00 00 07 fb	<D.....
Destination Port: 1161		
[Stream index: 0]		
[Stream Packet Number: 199]		
[Conversation completeness: Incomplete, DATA (15)]		
[TCP Segment Len: 0]		
Sequence Number: 1 (relative sequence number)		
Sequence Number (raw): 883061786		
[Next Sequence Number: 1 (relative sequence number)]		
Acknowledgment Number: 164091 (relative ack number)		
Acknowledgment number (raw): 232293103		
0101 = Header Length: 20 bytes (5)		
Flags: 0x010 (ACK)		
Window: 62780		
[Calculated window size: 62780]		
[Window size scaling factor: -2 (no window scaling used)]		
Checksum: 0x44a8 [unverified]		
Checksum Status: Unverified		
Urgent Pointer: 0		
[Timestamps]		
[SEQ/ACK analysis]		